PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Prognostic factors and prediction models for acute aortic dissection: a systematic review
AUTHORS	Ren, Yan; Huang, Shiyao; Li, Qianrui; Liu, Chunrong; Li, Ling; Tan, Jing; Zou, Kang; Sun, Xin

VERSION 1 – REVIEW

REVIEWER	Sarah R Haile, PhD Epidemiology, Biostatistics and Prevention Institute, University of Zurich, Switzerland
REVIEW RETURNED	17-Sep-2020

GENERAL COMMENTS	Thank you for the opportunity to review this manuscript. Overall the review seemed to be well done and the paper well written. Regarding the questionnaire: Are you aware of the TRIPOD statement and checklist? https://www.tripod-statement.org/ It would be interesting to see how well the various studies adhered to that. If you keep the discussion of your questionnaire, please include the items in an appendix. Generally, I think using a commonly used checklist like TRIPOD would be better than using a self-designed one. A review of the language would be helpful, as there are some strange phrases in some places.

REVIEWER	Spyridon Mylonas Department of Vascular and Endovascular Surgery University Hospital Cologne, Germany
REVIEW RETURNED	30-Oct-2020

GENERAL COMMENTS	The authors have conducted a review of the literature on the studies evaluating risk factors or predicting models for mortality in AAD patients. Hereby are my comments Is this review study submitted to a registry (i.e. Prospero)? Please make the appropriate statement. Abstract Section Data source and Methods are overlapping. Please remove Data source according the journal's instructions. Please remove the Strengths and Limitation Section form the

	Abstract, according to the Journal's instructions. "For the 14 prediction model studies" "For model development studies" Please replace the word " For" with " Among" Methods According to the PICOTS guide for systematic evaluation of prediction model performance the eligible studies should undergo a risk of bias test regarding the smaple size, the reporting/missing data, the statistical methodology (i.e. CHARMS checklist, or PROBAST model). Did the authors evaluate the risk of bias of the included studies? Please provide the relative diagram. References Moons KGM, de Groot JAH, Bouwmeester W, et al. Critical appraisal and data extraction for systematic reviews of prediction modelling studies: the CHARMS checklist. PLoS Med 2014;11:e1001744. doi:10.1371/journal.pmed.1001744. Wol R, Whiting P, Mallett S, et al. PROBAST: a risk of bias tool for prediction modelling studies. Cochrane Colloquium Vienna. 2015. Literature search and screening In case of duplicated publications of the same center on the same prognostic model with eventually the same patient sample were both of them included in your analysis, or the latest one? In the Appendix Table I seems to have been included both publications (i.g. Liu) Please clarify Results "…only six reported calibrations, and of which five reported good calibrations" What do you mean good calibrations? Please elaborate. "Thirty-one (97%) studies clearly described inclusion and exclusion criteria. All studies used consistent criteria and measurement of the studied population. For the outcome, all but one study13 used consistent criteria and measurement. For the analysis, 22 (69%) studies included all enrolled participants" The info provided in this paragraph is confusing. Please elaborate. Discussion A short comment of your findings about the robustness of the suggested prediction models and the current guidelines of the EACTS, ESVS, SVS ACTS could be of interest, as many of these societies have adopted these predicting models.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1:

Reviewer Name

Sarah R Haile, PhD

Institution and Country

Epidemiology, Biostatistics and Prevention Institute, University of Zurich, Switzerland

Please state any competing interests or state 'None declared':

None declared

Comments to the Author

Thank you for the opportunity to review this manuscript. Overall the review seemed to be well done and the paper well written.

Regarding the questionnaire: Are you aware of the TRIPOD statement and checklist? https://www.tripod-statement.org/ It would be interesting to see how well the various studies adhered to that. If you keep the discussion of your questionnaire, please include the items in an appendix. Generally, I think using a commonly used checklist like TRIPOD would be better than using a self-designed one.

Response: We fully appreciated your comments of this study. We understand that the TRIPOD checklist is used for assessing the reporting quality of prediction model studies. Since the aim of our study was to assess the methodological characteristics and the performance models, we designed the questionnaire by including items from PROBAST and CHARMS checklists. For clarification, we added our questionnaire as appendix B in this revision.

A review of the language would be helpful, as there are some strange phrases in some places.

Response: Thank you for your comments. We have carefully polished our language in the revision and we believe that the expressions are improved.

Reviewer 2:

Spyridon Mylonas

Institution and Country

Department of Vascular and Endovascular Surgery

University Hospital Cologne, Germany

Please state any competing interests or state 'None declared':

None declared

Comments to the Author

The authors have conducted a review of the literature on the studies evaluating risk factors or predicting models for mortality in AAD patients.

Hereby are my comments

Is this review study submitted to a registry (i.e. Prospero)? Please make the appropriate statement.

Response: Thank you for your inquiry. We confirm that this study was conducted according to a pre-specified protocol, but it was neither published nor registered.

Abstract
Section Data source and Methods are overlapping. Please remove Data source according the journal's instructions.

Please remove the Strengths and Limitation Section form the Abstract, according to the Journal's instructions.

"For the 14 prediction model studies"
"For model development studies"
Please replace the word " For" with " Among"

Response: Thank you for the very helpful comments. We have removed Data Source and Strengths and Limitations from the abstract, and changed the word 'For' to 'Among'.

Methods
According to the PICOTS guide for systematic evaluation of prediction model performance the eligible studies should undergo a risk of bias test regarding the smaple size, the reporting/missing data, the statistical methodology (i.e. CHARMS checklist, or PROBAST model). Did the authors evaluate the risk of bias of the included studies? Please provide the relative diagram.

References
Moons KGM, de Groot JAH, Bouwmeester W, et al. Critical appraisal and data extraction for systematic reviews of prediction modelling studies: the CHARMS checklist. PLoS Med 2014;11:e1001744. doi:10.1371/journal.pmed.1001744.
Wol R, Whiting P, Mallett S, et al. PROBAST: a risk of bias tool for prediction modelling studies. Cochrane Colloquium Vienna. 2015.

Response: Thank you for your comments. We added the assessment of risk of bias using the PROBAST tool in this revision. The detailed tool and assessment criteria were presented in Appendix C and the result were presented in Table 4.

Literature search and screening
In case of duplicated publications of the same center on the same prognostic model with eventually the same patient sample were both of them included in your analysis, or the latest one? In the Appendix Table I seems to have been included both publications (i.e. Liu) Please clarify

Response: We apologize for having not made the notes in the Appendix Table 1 that the Liu et al (2018a) study and the Liu et al (2018b) study are the different prognostic models in the previous version of the manuscript. Liu et al (2018a) study developed a prognostic model for the relationship between fibrinogen and in-hospital mortality in patients with type A acute aortic dissection, while Liu et al (2018b) study developed a prognostic mdel for the relationship between blood urea nitrogen and in-hospital mortality of patients with acute aortic dissection. We have added these clarification as footnotes in the Appendix Table 1.

Results
"…only six reported calibrations, and of which five reported good calibrations" What do you mean good calibrations? Please elaborate.

Response: Thank you for the very helpful comments. We considered good calibrations as a P value of Hosmer-Lemeshow test>0.05 and we added this criteria in paragraph 2, page 8.

"Thirty-one (97%) studies clearly described inclusion and exclusion criteria. All studies used consistent criteria and measurement of the studied population. For the outcome, all but one study13 used consistent criteria and measurement. For the analysis, 22 (69%) studies included all enrolled participants" The info provided in this paragraph is confusing. Please elaborate.

Response: Thank you for the very helpful comments. We have revised this paragraph to:

'Thirty-one (97%) studies clearly described inclusion and exclusion criteria for participants. The criteria used to define and to measure predictors in the study population were consistent among all included studies. The criteria for outcome (mortality) definition and measurement was consistent in all but one study13 (Table 3)'

Discussion
A short comment of your findings about the robustness of the suggested prediction models and the current guidelines of the EACTS, ESVS, SVS ACTS could be of interest, as many of these societies have adopted these predicting models.

Response: Thank you for your comments. In discussion, we added a short comment about the suggested prediction model as follows.

A prediction model most suitable for clinical practice should include a relatively small number of variables, be easily interpreted, and have good statistical performance. Apart from the well-established IRAD score model, our review found that the combined IRAD and CRP model used less variables and showed better discrimination than IRAD score alone. These characteristics may warrant daily practice of the combine model. Moreover, future studies may consider updating IRAD score by including other relevant biomarkers, which may further improve prognostic performance in clinical practice.

VERSION 2 – REVIEW

REVIEWER	Sarah R Haile Epidemiology, Biostatistics and Prevention Institute University of Zurich Switzerland
REVIEW RETURNED	15-Dec-2020

GENERAL COMMENTS	Thank you for your careful revision. Is it not possible to include the results of the TRIPOD checklist? What is the rationale for not using a previously developed checklist? Please describe in the manuscript. It is stated that calibration was evaluated using the Hosmer Lemeshow test. While this is a common test, various authors have argued that it is better to use other measures of calibration [A]. Please include another measure for calibration. [A] Van Calster, B., McLernon, D.J., van Smeden, M. et al. Calibration: the Achilles heel of predictive analytics. BMC Med 17, 230 (2019). https://doi.org/10.1186/s12916-019-1466-7